



Computational Chemistry Comparison and Benchmark DataBase

Release 22 (May 2022)

Standard Reference Database 101 [National Institute of Standards and Technology](#)

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Experimental data for O₂ (Oxygen diatomic)

22 02 02 11 45

Other names			
Dioxygen; Molecular oxygen; Oxygen; Oxygen molecule; Oxygen-16;			
INChI	INChIKey	SMILES	IUPAC name
InChI=1S/O2/c1-2	MYMOFIZGZYHOMD-UHFFFAOYSA-N	O=O	Dioxygen
State	Conformation		
$^3\Sigma_g$	D*H		

Enthalpy of formation (H_{fg}), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (C_p)

Property	Value	Uncertainty	units	Reference	Comment
H _{fg} (298.15K) $\Delta_f H^\circ$	0.00	0.00	kJ mol ⁻¹	CODATA	
H _{fg} (0K) $\Delta_f H^\circ$	0.00	0.00	kJ mol ⁻¹	CODATA	
Entropy (298.15K) S°	205.15	0.00	J K ⁻¹ mol ⁻¹	CODATA	
Integrated Heat Capacity (0 to 298.15K) $H^\circ - H^\circ(T)$	8.68	0.00	kJ mol ⁻¹	CODATA	
Heat Capacity (298.15K) C_p°	29.38		J K ⁻¹ mol ⁻¹	Gurvich	

Information can also be found for this species in the [NIST Chemistry Webbook](#)

Vibrational levels (cm⁻¹)

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm ⁻¹)	Harmonic(cm ⁻¹)	Reference	(km mol ⁻¹)	unc.	Reference		
1	Σ_g	1556	1580	1979HUB/HER					

Detailed diatomic data

ω_e	$\omega_e x_e$	$\omega_e y_e$	B_e	α_e	ZPE	reference
1580.161	11.95127	0.0458489	1.44562	0.0159305	787.3797	2007Iri:389

vibrational zero-point energy: 778.2 cm⁻¹ (from fundamental vibrations)

[Calculated vibrational frequencies](#) for O₂ (Oxygen diatomic).

More spectroscopic constants are available at the NIST Physics Laboratory website:

<http://physics.nist.gov/PhysRefData/MolSpec/Diatomic/Html/Tables/O2.html>

Rotational Constants (cm⁻¹)

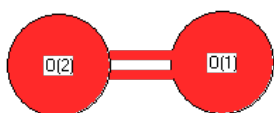
See section [I.F.4 to change rotational constant units](#)

A	B	C	reference	comment
	1.43768		1979HUB/HER	

[Calculated rotational constants](#) for O₂ (Oxygen diatomic).

Product of moments of inertia $I_A I_B I_C$		
11.72561	amu Å ²	1.947113E-39 gm cm ²

Geometric Data



Point Group $D_{\infty h}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rOO	1.208		1	2			1979HUB/HER	re

Cartesians

Atom	x (Å)	y (Å)	z (Å)
O1	0.0000	0.0000	0.0000
O2	0.0000	0.0000	1.2075

Atom - Atom Distances

Distances in Å

	O1	O2
O1		1.2075
O2	1.2075	

[Calculated geometries](#) for O₂ (Oxygen diatomic).

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
O=O	1

Connectivity

Atom 1	Atom 2
O1	O2

Electronic energy levels (cm⁻¹)

Energy (cm ⁻¹)	Degeneracy	reference	description
0	3	1979HUB/HER	$^3\Sigma_g$
7918.1	2	1979HUB/HER	$^1\Delta_g$
13195.1	1	1979HUB/HER	$^1\Sigma_g^+$
33057.3	1	1979HUB/HER	$^1\Sigma_u^-$

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
12.070	0.000	12.300		webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
0.451	0.007	webbook

Proton Affinity (kJ mol⁻¹)

Proton Affinity	unc.	Product	reference	comment
421.0		HO ₂ ⁺	webbook	

Dipole, Quadrupole and Polarizability**Electric dipole moment** →

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	³ Σ _g	D _{∞h}	True				0.000	NSRDS-NBS10		D _{∞h}	0	1
2	1	¹ Δ _g	D _{∞h}	False							D _{∞h}	0	1
3	1	¹ Σ _g ⁺	D _{∞h}	False							D _{∞h}	0	1
4	1	¹ Σ _u ⁻	D _{∞h}	False							D _{∞h}	0	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

[Calculated electric dipole moments](#) for O₂ (Oxygen diatomic).

Electric quadrupole moment +₂⁻

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	³ Σ _g	D _{∞h}	True	0.155	0.155	-0.310	2014Cou/Nto:41	⁺⁻ 0.0080944	D _{∞h}	0	1
2	1	¹ Δ _g	D _{∞h}	False						D _{∞h}	0	1
3	1	¹ Σ _g ⁺	D _{∞h}	False						D _{∞h}	0	1
4	1	¹ Σ _u ⁻	D _{∞h}	False						D _{∞h}	0	1

[Calculated electric quadrupole moments](#) for O₂ (Oxygen diatomic).

Electric dipole polarizability (Å³) ↔

alpha	unc.	Reference
1.562		1997Oln/Can:59

[Calculated electric dipole polarizability](#) for O₂ (Oxygen diatomic).

References

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squib	reference	DOI
1979HUB/HER	Huber, K.P.; Herzberg, G., Molecular Spectra and Molecular Structure. IV. Constants of Diatomic Molecules, Van Nostrand Reinhold Co., 1979	10.1007/978-1-4757-0961-2
1997Oln/Can:59	TN Olney, NM Cann, G Cooper, CE Brion, Absolute scale determination for photoabsorption spectra and the calculation of molecular properties using dipole sum-rules, Chem. Phys. 223 (1997) 59-98	10.1016/S0301-0104(97)00145-6
2007Iri:389	KK Irikura "Experimental Vibrational Zero-Point Energies: Diatomic Molecules" J. Phys. Chem. Ref. Data 36(2), 389, 2007	10.1063/1.2436891
2014Cou/Nto:41	VW Couling, SS Ntombela "The electric quadrupole moment of O ₂ " Chem. Phys. Lett. 614 (2014) 41-44	10.1016/j.cplett.2014.09.006
CODATA	Cox, J.D.; Wagman, D.D.; Medvedev, V.A.CODATA Key Values for Thermodynamics. Hemisphere, New York, 1989	10.1002/bbpc.19900940121
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	

NSRDS-NBS10	R. D. Nelson Jr., D. R. Lide, A. A. Maryott "Selected Values of electric dipole moments for molecules in the gas phase" NSRDS-NBS10, 1967	10.6028/NBS.NSRDS.10
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303
Got a better number? Please email us at cccbdb@nist.gov		

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AsSe (Arsenic monoselenide)	O2- (oxygen diatomic anion)